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L-Lysine – Coupling of Bioreaction and Process Model

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7.1 Introduction

To date, optimization of chemical and biochemical processes is achieved by empirical or intuitive methods. Industrial research in particular is characterized by enormous pressure of time and costs. There is a need for a systematic method to evaluate whether or not expense of research in process optimization will be compensated for by a possible reduction in production costs. This creates a demand for methods that enable an estimation of the essential economic data of a process to be made in order to develop decision criteria for the general research strategy. Additionally, there is a need for fast and unerring methods for the determination of process optima.

Biochemical processes in particular are characterized by a large number of influential parameters and their variations. Empirical, not fully systematic, strategies are time- and cost-intensive. In this context, mathematical modeling of biotechnological systems has proven to be an efficient and modern tool to evaluate these bioprocesses. This case study presents methods for bioprocess development by modeling in an industrial environment and examines the advantages of this approach. It combines a dynamic model of the bioconversion, the core part of each bioprocess, with a model of the overall production process.

The production of L-lysine, a well established bioprocess, is used as a case study. Lysine is mainly used as an animal feed additive. The annual production is approximately 700 000 tons/year. A comprehensive overview of lysine production is given by Pfefferle et al. [7.1].

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7.2 Basic Strategy

The strategy is based on the idea of coupling a dynamic model describing the bioreaction to a process model of the complete production plant. Because the reaction is integrated in a complex network of unit procedures, the model must consequently be capable of calculating the energy and mass balances as well as the cost factors of the entire process.

Whereas the first part can be performed by almost any numerical simulation tool, the second part is ideally carried out using a process-simulation tool. For the bioreaction model, considering the kinetics of the lysine production, Modelmaker® (Modelmaker Tools BV, Arnhem, The Netherlands) was used, while the process model was built using SuperPro Designer®. The aim of this strategy is to achieve information about what impact a process parameter of the bioreaction has on the economic or environmental objective functions of the complete production plant. This overall information is of greater interest than the optimization of a single unit procedure. The numerical simulation of the bioreaction alone gives no information about the impact that parameter variations have on the nature and economic efficiency of the overall process. The process model, however, requires data about the time and stoichiometry of the bioreaction. This information can be derived from experimental results or, as in this study, from simulation results.

7.3 Bioreaction Model

The first fundamental step is a mathematical description of the biological activities. Basic data usually emerge from a few preliminary experiments. The resulting model should be able to reflect the influence of relevant process parameters with adequate accuracy. A procedure serving to establish such a model is described by Büchs [7.2]. The present study makes use of an existing biological model [7.2] that is described in detail in the Appendix, and model parameters are presented in Table 7.A1. This model describes the production of the amino acid lysine in a fed-batch process by a mutant of the bacterium *Corynebacterium glutamicum*.

In wild type strains of *C. glutamicum*, lysine is usually feedback regulated by the amino acid threonine. Therefore it is assumed that the applied mutant is disrupted in threonine metabolism, resulting in a threonine auxotrophy. Even though such amino acid-auxotrophic mutants have disappeared from industrial lysine production [7.1], this simplified study is based on such a strain. The reason is that this particular strain characteristic is very suitable to explain the introduced optimization approach. Threonine is an essential component of proteins and has to be supplemented to the culture media in case of auxotrophic strains. Therefore the carbon flux into cell growth and lysine production, and further the stoichiometry of the applied microorganism's metabolism, can be controlled by the medium-specific process parameter 'initial threonine concentration' ($c_{\mathrm{Thr_IN}}$) [7.2]. Figure 7.1 shows three different pathways which carbon may take within general metabolism. Variation of these three carbon fluxes will alter the substrate yield.

In Equation (7.1) the glucose-consumption rates for lysine, growth, and cell maintenance are taken into account. The fluxes are characterized by the corresponding coefficients and

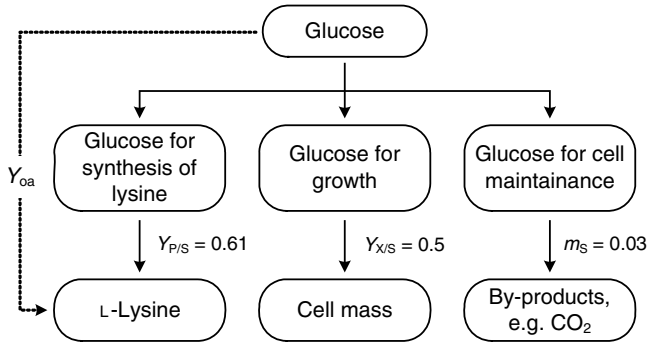


Figure 7.1 *Stoichiometry of glucose consumption for a lysine-producing strain of* C. glutamicum *[7.2]. For abbreviations see Nomenclature section. Reproduced by permission of Wiley-VCH*

describe the stoichiometry of the glucose consumption of lysine-producing microorganisms. For abbreviations see the Nomenclature section.

$$\frac{dc_S}{dt} = -\frac{1}{Y_{X/S}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/S}} \cdot r_P \cdot c_X - m_S \cdot c_X \tag{7.1}$$

Once the biological model is established and validated, it can be used to simulate and optimize the bioreaction. Figure 7.2 shows a simulation of a fed-batch process of lysine production with an initial threonine concentration of 1.6 g/L as an example. After depletion of the glucose in the initial batch phase, a fed-batch phase is initiated where the feed rate is controlled by the dissolved oxygen value (DO_2). A DO_2 set point of 25% of air saturation was chosen. Owing to the limited oxygen-transfer capacities of conventional fermenters, the oxygen transfer limits the feed rate of fed-batch fermentations and therefore might extend the fermentation time [7.3, 7.4].

To account for a limited oxygen transfer in the fermenter, the start k_La value was assumed to be 1000 h^{-1}. The maximal oxygen-transfer rate at a dissolved-oxygen value of 25% is limited to a value of around 0.2 mol/L h. Figure 7.2 depicts the course of threonine, biomass, and lysine concentration during the process. As long as threonine is present in the medium, cell growth is favored, while only small amounts of lysine are produced. After the complete consumption of threonine, cell growth stops and lysine synthesis is enhanced. The biomass concentration decreases due to the dilution of the fermentation broth by the substrate feed stream. An important problem is specifying the optimization function. Figure 7.3 recapitulates the most influential input (left) and output parameters (right) for this process. The input parameters can be classified into three categories:

1. Strain-specific parameter, e.g. maximum specific growth rate, specific productivity, etc.
2. Medium-specific parameter, e.g. concentration of the main carbon source glucose and the initial threonine concentration.
3. Operation-specific parameter, e.g. dilution rate of a continuous fermentation, feed rate of a fed-batch fermentation.

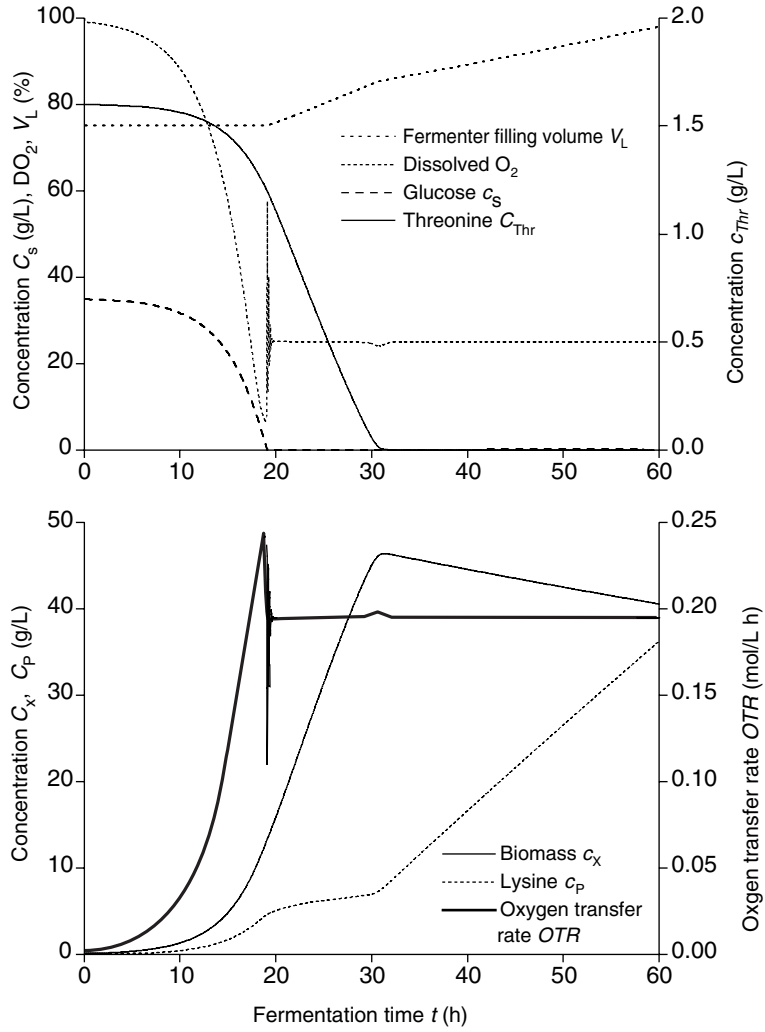


Figure 7.2 *Simulated kinetics of c_S, V_L, DO_2, OTR, c_P, c_X, c_{Thr} versus fermentation time for an example of an initial threonine concentration of 1.6 g/L for a process of lysine production in a glucose-fed-batch of 700 g/L substrate concentration in the feed*

The most important output parameters are the space-time yield *STY*, the overall yield Y_{oa}, and the product concentration c_P. These parameters can be obtained from the simulations of the bioreaction model. In this work, only three optimization parameters are considered. The question arises as to which of these parameters should be optimized. A purely intuitive optimization of the complete system is extremely difficult if not impossible to achieve.

To demonstrate this challenge, the initial threonine concentration c_{Thr_IN} as a medium-specific parameter may be chosen for variation. Figure 7.4a shows the final biomass concentration c_{X_END}, the final lysine concentration c_{P_END}, and the total fermentation time t_{END}

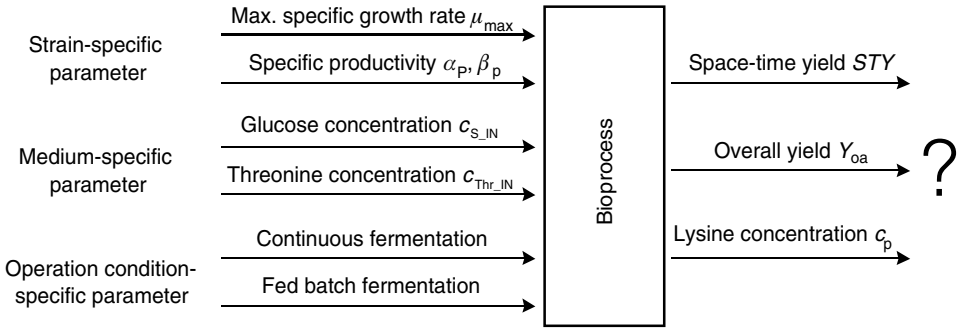


Figure 7.3 *Problem of assessment criteria for bioprocess optimization for the example of a lysine-producing strain of* C. glutamicum. *Reproduced by permission of Wiley-VCH*

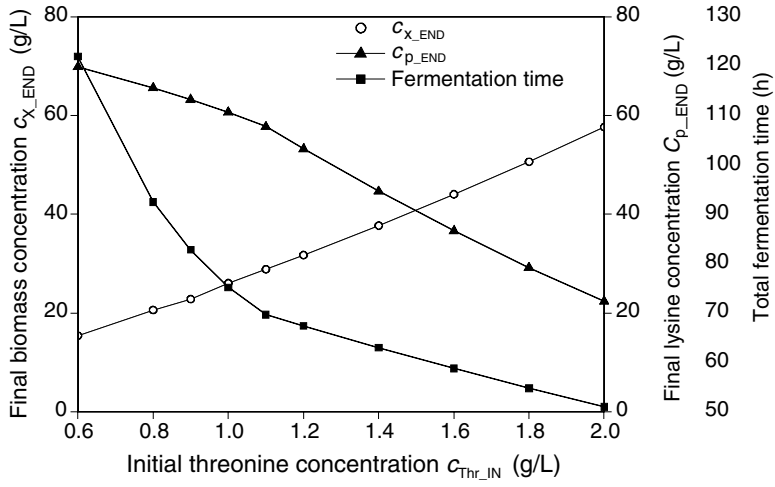
in the bioreaction model versus initial threonine concentration c_{Thr_IN} from 0.6 to 2.0 g/L. The final biomass concentration increases with increasing c_{Thr_IN}. The lysine concentration decreases because more glucose is consumed for the growth of biomass. Finally, the total fermentation time, being a very important process parameter with a strong influence on the process economy, decreases with increasing initial threonine concentrations.

In Figure 7.4b, the output parameters space-time yield and overall yield are plotted versus initial threonine concentration. The c_{Thr_IN} values for maximal values of these two parameters are not at all equal. This means that the highest Y_{oa} cannot be achieved together with the highest final space-time yield.

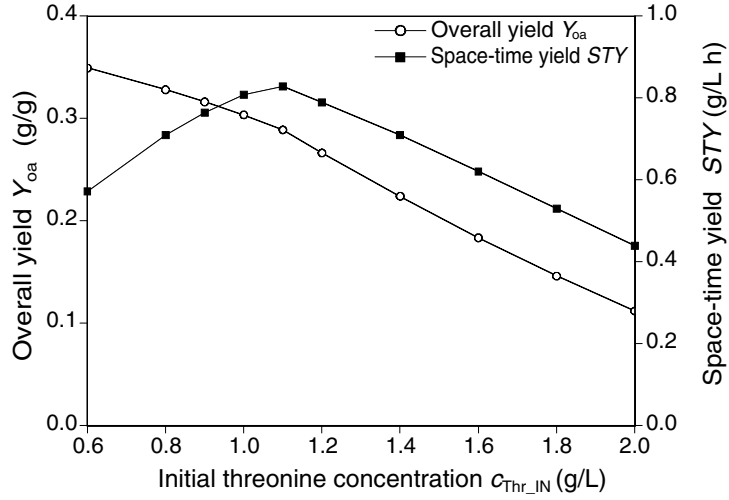
7.4 Process Model

The process model of an industrial L-lysine process is derived from Stevens [7.5]. It should be noted that details of recent lysine-production processes are neither published nor distributed by the lysine-producing companies. Therefore, this example represents neither the actual sequence of unit procedures nor the real manufacturing costs of the final product. The process flow diagram is shown in Figure 7.5.

The flow sheet consists of the fermentation and the downstream section. Since the desired final product is supposed to be used as a feed additive, the downstream section is kept very simple [7.1, 7.5]. In the first part of the process the required amounts of water, threonine, and nutrients are mixed in a blending tank (P-1, units of stirred tanks of maximum capacity 80 m^3). Additional water is added separately by a mixing unit (P-8) to lower the cost for P-1. As nutrients, only the C-source (glucose) and the P-source (KH_2PO_4) are considered. The N-source, NH_4OH, will be titrated into the fermenter during the reaction. The culture medium is sterilized through a heat sterilizer (P-2) passing the medium on to the fermenter unit (P-20) (units of stirred tank fermenters of maximum capacity 300 m^3). Since SuperPro Designer® accounts for a fixed stoichiometry of a reaction and a fixed total fermentation time, the fed-batch process is modeled as a batch process with an initial glucose concentration of 200 g/L. The mass balance of the elements C, O, P, and N in the bioreactor is derived from the stoichiometry from the bioreaction model and the elemental analysis of



(a)



(b)

Figure 7.4 *(a) Final biomass concentration and lysine concentration, and (b) overall yield and final space-time yield for initial threonine concentrations from 0.6 to 2.0 g/L for a process of lysine production with* C. glutamicum *in a fed-batch mode*

biomass and lysine. After the fermentation, the fermentation broth is transferred to a storage vessel (P-6). From this vessel, the broth is transferred to a rotary vacuum filter (P-10) to separate the biomass from the liquid. The permeate is transferred to an evaporation unit (P-23) that removes around 80% of the water content of the liquid stream. After storage in a vessel (P-12) the broth is spray dried and processed to granules in P-15. These granules are transferred to a storage tank (P-17).

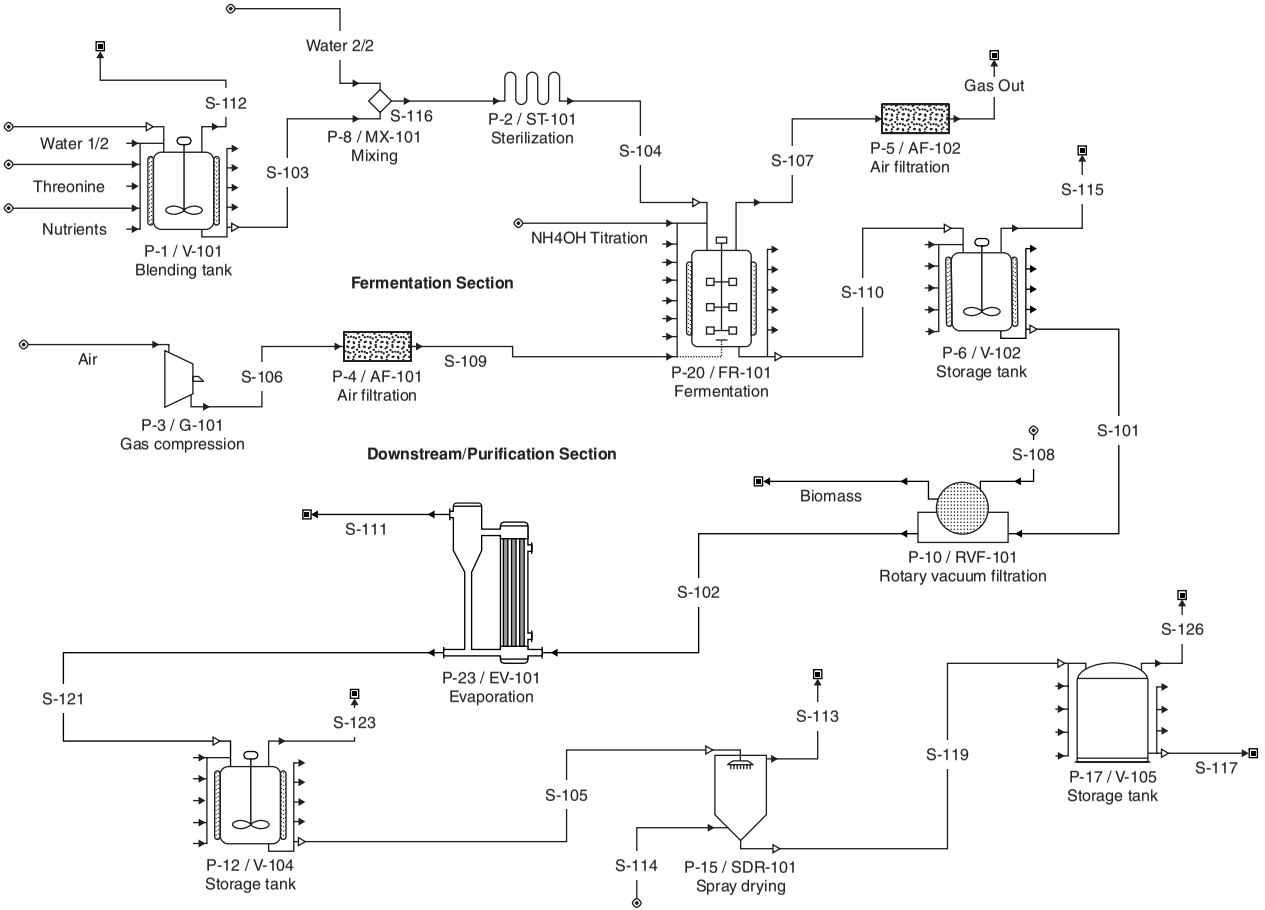


Figure 7.5 *Process flow diagram of a lysine-production plant*

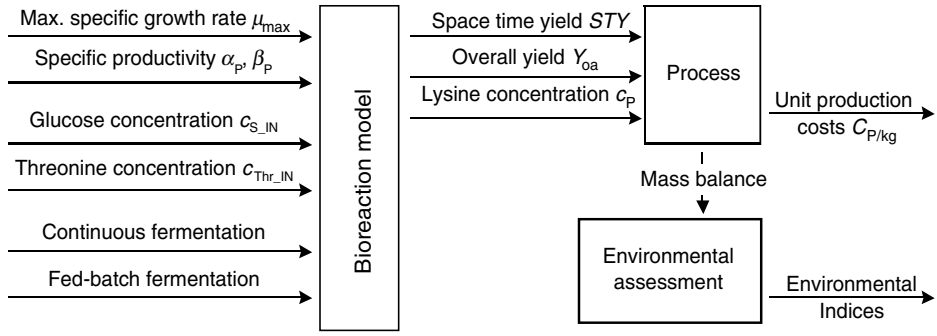


Figure 7.6 *Assessment criterion for bioprocess optimization derived from coupling the bioreaction model and process model of a lysine-producing strain of* C. glutamicum. *The reaction model is simulated with Modelmaker®-software; the process model is implemented in the simulation software SuperPro Designer®*

7.5 Coupling of Bioreaction and Process Model

An optimization of the complete process based only on the biochemical reaction model is insufficient. Figure 7.6 represents a general method to solve the problem of optimizing the complete process. The bioreaction model is coupled to the process model created in SuperPro Designer®.

In this case study, the bioreaction model as well as the process model are used as standalone systems with different features. The bioreaction model contributes information about the stoichiometry and time behavior of the biological reactions. The process model is able to calculate the energy and mass balances and the economic behavior (among many other parameters) for a complete production plant of a fixed reaction stoichiometry and fixed reaction time. Subsequently, the mass balance that is calculated in the process model can also be used for the environmental assessment. To investigate the influence of different input parameters of the bioreaction model on the process economic behavior (i.e. unit production costs) or on the environmental impact, the output parameters of the bioreaction model are transferred to the process model. For every c_{Thr_IN}, the stoichiometry and total fermentation time (t_{END}) are calculated with the bioreaction model. A mass-based stoichiometry of the presented process of lysine production derived from the bioreaction model is demonstrated in Equation (7.2) for the example of $c_{Thr_IN} = 1.1$ g/L.

$$\begin{aligned} &200\text{ g Glucose} + 3.8\text{ g } KH_2PO_4 + 37.8\text{ g } NH_4OH + 1.1\text{ g Threonine} \rightarrow \\ &28.9\text{ g Biomass} + 57.7\text{ g Lysine} + 30\text{ g Water} + 126.1\text{g } CO_2 \end{aligned} \tag{7.2}$$

These parameters are used in the process model of the production plant. From this, production costs or environmental impact can be calculated. The calculations are conducted following a user-defined stoichiometry of the biochemical or chemical reactions as well as a user defined total fermentation time. A variation of a process parameter (such as c_{Thr_IN}) that effects the stoichiometry and therefore the amount of product per batch, as well as the

total fermentation time, will determine the mass output of the plant. Since a production plant is usually designed for a certain amount of product per year, this has to be considered for comparative studies or scenario analysis. If the equipment size of the unit procedures is not adapted to stoichiometric changes of the process, the annual process throughput will change. To maintain a constant annual production, the throughput-adjustment tool of SuperPro Designer® is applied to automatically re-adapt the annual production with varying bioreaction variables. A higher product titer, for example, would result in a smaller size and, therefore, lower cost for the fermentation unit. For the present example, the annual process throughput was fixed to 6000 metric tons of lysine. This value represents around 1% of the estimated worldwide demand of lysine per year [7.6]. The optimization procedure is now demonstrated for the above discussed process model of a lysine-production plant.

7.5.1 Assumptions

(i) Type and sequence of unit procedures are considered as known and constant in this example. The process simulator allows this to be altered if desired.
(ii) For simplification, trace elements and other nutrients are not considered, but only C, N, and P assimilation for biomass and lysine synthesis are taken into account.
(iii) Product losses of the different unit procedures in the downstream processing are considered as constant.

For the biomass a C-content of 50% w/w is assumed. The P- and N-content of biomass of *C. glutamicum* is around 3% w/w and 14% w/w, respectively [7.7]. For N consumption, the N-content of lysine is also considered. One mole of lysine contains two moles of N. Nitrogen is provided as NH_4OH with a molecular weight of 35 g/mol. The molecular weight of lysine is 146 g/mol. Hence, for the synthesis of 1 g of lysine, 0.48 g of NH_4OH is required.

As a result of the bioreaction model, the reaction stoichiometry for a certain initial threonine concentration, as demonstrated in Equation (7.2), has to be adjusted in the Reaction Stoichiometry module of the process model. This stoichiometric equation is part of the reaction operation within the unit procedure Fermentation. Starting from a base case process model, several new scenarios were created, each including the resulting reaction stoichiometry for initial threonine concentrations from 0.6 to 2.0 g/L in steps of 0.2 g/L. This led to eight different scenarios of the same plant, which were used for the economic and environmental assessment. Having adjusted the annual process throughput to 6000 metric tons per year, the process model can be solved. Mass and energy balances for the process are solved and the required equipment size of each unit procedure is calculated.

Before deriving the information about economic and environmental impacts of each scenario, another aspect has to be considered. Regardless of the amount of lysine or biomass to be produced, glucose is assumed to be completely consumed during fermentation. However, the amount of lysine and biomass determines the amount of N- and P-source consumed during fermentation. The amount of N- and P-source influences both the economic calculations and the environmental assessment. The addition of NH_4OH and KH_2PO_4 was adjusted such that a residual amount of 5 kg per fermentation was registered in every batch. To enable the comparison of each scenario, the residual values have to be equalized. After properly adjusting these parameters, the mass and energy balances can be solved again.

7.6 Results and Discussion

For the environmental assessment, only the Environmental index EI_{Mult} is applied for the output components in this study. Components under consideration are glucose, threonine, NH_4OH, KH_2PO_4, biomass, lysine loss, and CO_2. The EI_{Mult}, $C_{P/kg}$, *STY* and Y_{oa} values are plotted versus the optimization parameter c_{Thr_IN} in Figure 7.7.

Figure 7.7 represents the results of coupling the bioreaction model to the process model. In addition to overall yield and space-time yield, the production costs and EI_{Mult} Output is plotted versus initial threonine concentration. The unit-production cost decreases to a minimum of \$ 4.9/kg L-lysine at an initial threonine concentration of 0.9 g/L and rises with increasing initial threonine concentrations up to \$ 12/kg L-lysine at c_{Thr_IN} of 2.0 g/L.

Within the investigated range of initial threonine concentrations, the EI_{Mult} rises continuously from 4.1 index points/kg P to 13.1 index points/kg P. The increasing *EI* with increasing c_{Thr_IN} is mainly caused by an increase of biomass and CO_2 at higher c_{Thr_IN}, which are considered as waste compounds. The more that glucose is converted into biomass and CO_2 instead of lysine, the higher is the EI_{Mult}. The difference between the minimum EI_{Mult} Output at 0.6 g/L threonine and the EI_{Mult} Output at 0.9 g/L where the unit-production cost shows its minimum is very small. Thus, running the process under the economically optimal conditions also results in an environmental impact near the minimum.

As seen in Figure 7.7, the minimal production costs are not achieved at the same c_{Thr_IN} as the highest final space-time yield (*STY*) or overall lysine yield (Y_{oa}). It can be concluded that overall yield or space-time yield are not the most appropriate process parameters to be optimized in this case study.

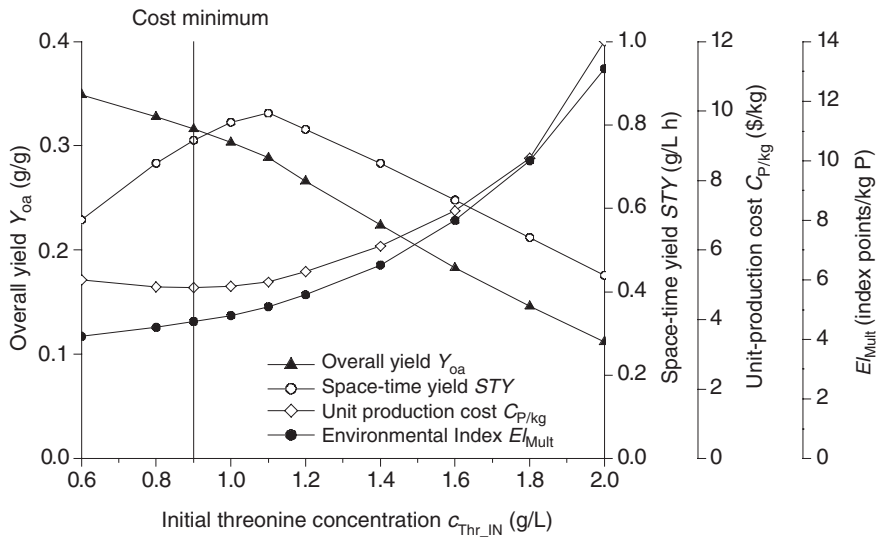


Figure 7.7 *Overall yield, final space-time yield, unit-production cost, and environmental index (EI_{Mult} Output) versus the initial threonine concentration from 0.6 to 2.0 g/L for a process of lysine production with* C. glutamicum *in a fed-batch mode*

Owing to the application of a threonine auxotroph mutant of *C. glutamicum,* the achieved minimal production costs of $ 4.9/kg L-lysine might no longer be the state of the art. Regarding the limited knowledge used in this case equivalent to an early stage of process development, however, the achieved production costs coincide quite well with real industrial prices. While Festel et al. [7.8] published production cost of $ 1–2/kg L-lysine for an unspecified sequence of unit procedures and quality of product, the calculated value from this example is in the same order of magnitude.

Suggested Exercises

1. The purchase equipment cost of the process is calculated using the built-in model of SuperPro Designer®. Naturally, this provides only a first estimate. For a bioreactor (P-20), you have obtained a quotation of $ 2 million (FoB) for the desired volume of 550 m^3. Assume that the transport of a reactor to the facility costs $ 150 000. Set the new price (Equipment Data of P-20) and study how capital investment and operating costs vary.
2. The first step of the downstream processing is the removal of the biomass. In the base case, rotary vacuum filtration is assumed. Model an alternative process setting with a disk-stack centrifuge that removes 98% of the biomass and a subsequent ultrafiltration that retains the remaining biomass. First, use the default values for the parameters in the newly added unit operations. Compare the economic performance with the base case. Then, go through the two unit models and check if the default values are realistic for this case. Make reasonable changes and watch the effect on the process performance.

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Nomenclature

c_L = oxygen concentration (g/L)
c_P = product concentration (g/L)
c_{P_END} = final product concentration (g/L)
$C_{P/kg}$ = unit-production cost ($/kg)
c_S = substrate concentration (g/L)
c_{SF} = substrate concentration in the feed (g/L)
c_{S_IN} = initial substrate concentration (g/L)
c_{Thr} = threonine concentration (g/L)
c_{Thr_IN} = initial threonine concentration (g/L)
c_X = biomass concentration (g/L)
c_{X_END} = final biomass concentration (g/L)
c_{X_IN} = initial biomass concentration (g/L)
DO_2 = dissolved oxygen (%)
EI_{Mult} = Environmental Index (index points/kg P)
F = rate of feed (feed rate) (L/h) or (m^3/h)
K_La = specific mass transfer coefficient (1/h)
K_{IP} = product inhibition constant (g/L)
K_{IThr} = threonine inhibition constant (g/L)
K_O = substrate oxygen affinity constant (g/L)
K_{PS} = product affinity constant (g/L)
K_S = substrate carbon source affinity constant (g/L)
K_{Thr} = substrate threonine affinity constant (g/L)
L_{O_2} = oxygen solubility (mol/L/bar)
m_O = specific oxygen consumption for maintenance (g/L)
m_S = specific substrate consumption for maintenance (g/L)
OTR = oxygen transfer rate (mol/L h)
P_R = reactor pressure (bar)
r_P = rate of lysine production (g/L h)
STY = space time yield (g/L h)
t = time
t_{END} = total fermentation time (h)
V = fermenter filling volume (m^3)
V_L = fermenter filling volume (%)
y_L = mole fraction of oxygen in the liquid phase (mol/mol)
y_{O_2} = mole fraction of oxygen in the gas phase (mol/mol)
Y_{oa} = overall yield (g/g)
$Y_{P/O}$ = product yield per amount of oxygen (g/g)
$Y_{P/S}$ = product yield per amount of substrate (g/g)
$Y_{X/S}$ = biomass yield per amount of substrate (g/g)
$Y_{X/O}$ = biomass yield per amount of oxygen (g/g)
$Y_{X/Thr}$ = biomass yield per amount of threonine (g/g)
α_P = growth-associated coefficient for product synthesis (g/g)
β_P = non-growth-associated coefficient for product synthesis (g/g h)
μ = specific growth rate (1/h)
μ_{max} = maximum specific growth rate (1/h)

Appendix 7.1 Biochemical Reaction Model for a Fed-Batch Fermentation to Produce Lysine

Mass balance for glucose

$$\frac{dc_S}{dt} = -\frac{1}{Y_{X/S}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/S}} \cdot r_P \cdot c_X - m_S \cdot c_X + \frac{F}{V}(c_{SF} - c_S)$$

Mass balance for oxygen $\frac{dc_L}{dt} = -\frac{1}{Y_{X/O}} \cdot \mu \cdot c_X - \frac{1}{Y_{P/O}} \cdot r_P \cdot c_X - m_O \cdot c_X$

Mass balance for threonine $\frac{dc_{Thr}}{dt} = -\frac{1}{Y_{X/Thr}} \cdot \mu \cdot c_X - \frac{F}{V} \cdot c_{Thr}$

Mass balance for biomass $\frac{dc_X}{dt} = \mu \cdot c_X - \frac{F}{V} \cdot c_X$

Mass balance for lysine $\frac{dc_P}{dt} = r_P \cdot c_X - \frac{F}{V} \cdot c_P$

Mass balance for the fermenter volume $\frac{dV}{dt} = F$

Kinetic model for oxygen transfer $OTR = k_L a \cdot L_{O_2} \cdot p_R \cdot (y_{O_2} - y_L)$

Kinetic model for growth $\mu = \mu_{max} \frac{c_S}{c_S + K_S} \cdot \frac{c_L}{c_L + K_O} \cdot \frac{c_{Thr}}{c_{Thr} + K_{Thr}}$

Kinetic model for lysine formation

$$r_P = (\alpha_P \cdot \mu + \beta_P) \cdot \frac{c_S}{c_S + K_{PS}} \cdot \frac{c_L}{c_L + K_O} \cdot \frac{K_{IThr}}{c_{Thr} + K_{IThr}} \cdot \frac{K_{IP}}{c_P + K_{IP}}$$

Overall yield $Y_{oa} = \frac{c_P}{c_{S_IN}}$

Space-time yield $STY = \frac{c_P}{t}$

Table 7.A1 *Parameters of the biological model taken from [7.2]*

Stoichiometric constants	Value	Unit
$Y_{X/S}$	0.52	(g/g)
$Y_{P/S}$	0.6	(g/g)
$Y_{X/Thr}$	33	(g/g)
$Y_{P/O}$	4.11	(g/g)
$Y_{X/O}$	1.29	(g/g)
Monod constants		
μ_{max}	0.28	(1/h)
K_O	6.4×10^{-6}	(g/L)
K_S	0.1	(g/L)
K_{Thr}	0.1	(g/L)
K_{PS}	0.072	(g/L)
L_{O_2}	0.00118	(mol/L/bar)
m_O	0.036	(g/L)
m_S	0.034	(g/L)
α	0.2	(g/g)
β	0.043	(g/g h)
c_{S_IN}	50	(g/L)
c_{X_IN}	0.1	(g/L)